

MUCKLE HOLM, United Kingdom

WMO: 030070

Lat: 60.581N

Lon: 1.265W

Elev: 66

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 90-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.7	32.4	20.5	15.4	34.6	23.2	17.5	35.2	55.2	42.7	51.1	40.5	20.2	320	0.868

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	60.1	57.1	58.5	56.0	57.4	55.1	58.0	59.4	56.7	58.0	55.7	57.0	11.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.0	69.5	58.9	55.9	66.6	57.8	54.8	64.0	56.7	25.1	59.7	24.3	58.1	23.6	57.1	65.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
46.3	40.9	35.2	DB	27.2	64.1	5.1	2.1	23.6	65.6	20.6	66.8	17.7	68.0	14.0	69.5
			WB	24.9	60.4	4.2	2.6	21.8	62.3	19.4	63.9	17.0	65.3	13.9	67.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	42.0	40.7	41.4	42.8	45.0	48.9	52.6	54.7	52.1	48.3	44.9	42.0
	DBStd	5.84	3.27	3.87	3.57	3.29	4.97	3.13	3.19	2.28	2.80	3.42	3.41	4.01
	HDD50	1699	247	259	267	216	159	54	12	0	12	72	154	247
	HDD65	6814	712	679	732	666	619	483	385	318	387	518	603	712
	CDD50	361	0	0	0	0	5	21	92	148	76	18	1	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.3	19.6	21.4	18.8	15.8	12.8	13.4	11.7	11.6	14.8	18.6	19.1	17.9
Precipitation	PrecAvg	45.4	5.7	4.4	4.2	2.5	2.3	2.2	2.4	3.2	3.6	4.5	5.1	5.2
	PrecMax	54.9	10.9	6.9	7.3	4.2	4.8	4.3	5.2	6.3	5.5	7.2	7.7	9.2
	PrecMin	38.6	1.8	2.0	1.5	0.9	0.4	0.4	0.9	1.5	1.1	1.9	2.3	2.0
	PrecStd	4.6	2.0	1.2	1.1	0.7	0.9	0.9	1.0	1.1	1.1	1.3	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	50.4	49.8	51.4	58.5	59.6	62.2	62.6	60.0	55.7	53.1	51.6
		MCWB	47.2	48.7	47.6	48.1	52.4	53.6	58.8	59.8	57.1	53.9	51.5	49.7
	2%	DB	48.5	49.0	48.4	49.5	53.8	55.6	59.3	60.6	57.9	54.2	51.5	50.0
		MCWB	46.4	47.3	46.3	47.1	50.3	51.9	56.7	58.1	55.7	52.4	49.7	48.5
	5%	DB	47.6	47.5	47.3	48.1	51.4	53.8	57.8	59.1	56.6	53.4	50.5	48.8
		MCWB	45.6	45.6	45.3	46.3	49.0	51.1	55.5	56.7	54.7	51.7	48.9	47.2
	10%	DB	46.6	46.2	46.3	46.9	49.9	52.7	56.6	57.8	55.7	52.7	49.7	47.6
		MCWB	44.8	44.2	44.2	45.0	47.7	50.4	54.4	55.5	53.8	50.9	48.1	45.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	49.5	48.5	49.4	53.4	54.5	59.4	60.6	57.7	54.7	52.1	50.3
		MCDB	49.0	49.9	49.2	50.3	57.4	57.2	61.7	62.1	58.9	55.1	52.7	50.9
	2%	WB	47.2	47.7	46.9	48.0	51.0	52.9	57.4	58.5	56.5	53.1	50.5	49.0
		MCDB	47.9	48.7	47.7	49.0	53.2	54.8	58.8	60.0	57.5	53.8	51.2	49.8
	5%	WB	46.2	45.9	45.9	46.6	49.6	51.9	56.0	57.3	55.3	52.2	49.6	47.8
		MCDB	47.1	46.9	46.8	47.4	51.2	53.2	57.4	58.6	56.4	53.1	50.3	48.7
	10%	WB	45.1	44.5	44.7	45.5	48.3	51.1	54.8	56.2	54.2	51.3	48.5	46.2
		MCDB	46.3	46.0	45.8	46.5	49.5	52.3	56.2	57.5	55.2	52.5	49.5	47.2
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.1	5.0	4.8	4.5	4.1	4.4	4.4	3.9	4.0	4.3	4.8
		MCDBR	5.3	6.2	5.8	6.2	6.5	5.8	6.4	6.3	5.0	4.2	5.1	5.0
		MCWBR	5.8	6.3	5.8	5.2	5.5	4.8	5.3	5.4	4.3	4.6	6.0	5.5
	5% WB	MCDBR	5.3	6.3	5.8	5.5	6.1	5.3	6.5	6.2	4.8	4.4	5.2	4.8
		MCWBR	6.2	6.9	6.1	5.2	5.5	4.8	5.6	5.6	4.2	5.1	6.3	5.7
Clear-Sky Solar Irradiance	taub	0.334	0.355	0.369	0.383	0.373	0.366	0.379	0.366	0.363	0.346	0.328	0.407	
	taud	2.075	2.194	2.269	2.298	2.394	2.455	2.434	2.471	2.437	2.317	2.130	2.376	
	Ebn at Noon	117	193	236	257	270	274	267	262	239	196	119	76	
	Edh at Noon	15	23	29	34	33	32	32	29	25	20	14	10	
All-Sky Solar Radiation	RadAvg	69	226	566	1060	1479	1534	1382	1125	679	339	102	43	
	RadStd	7	25	41	62	94	166	121	108	62	35	10	3	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (19 neighbors)	+0.63	+1.21	+1.26	N/A	N/A	N/A	-184	-241	N/A	N/A	

Nomenclature: See separate page